



Global Wine Review & Evaluation Database

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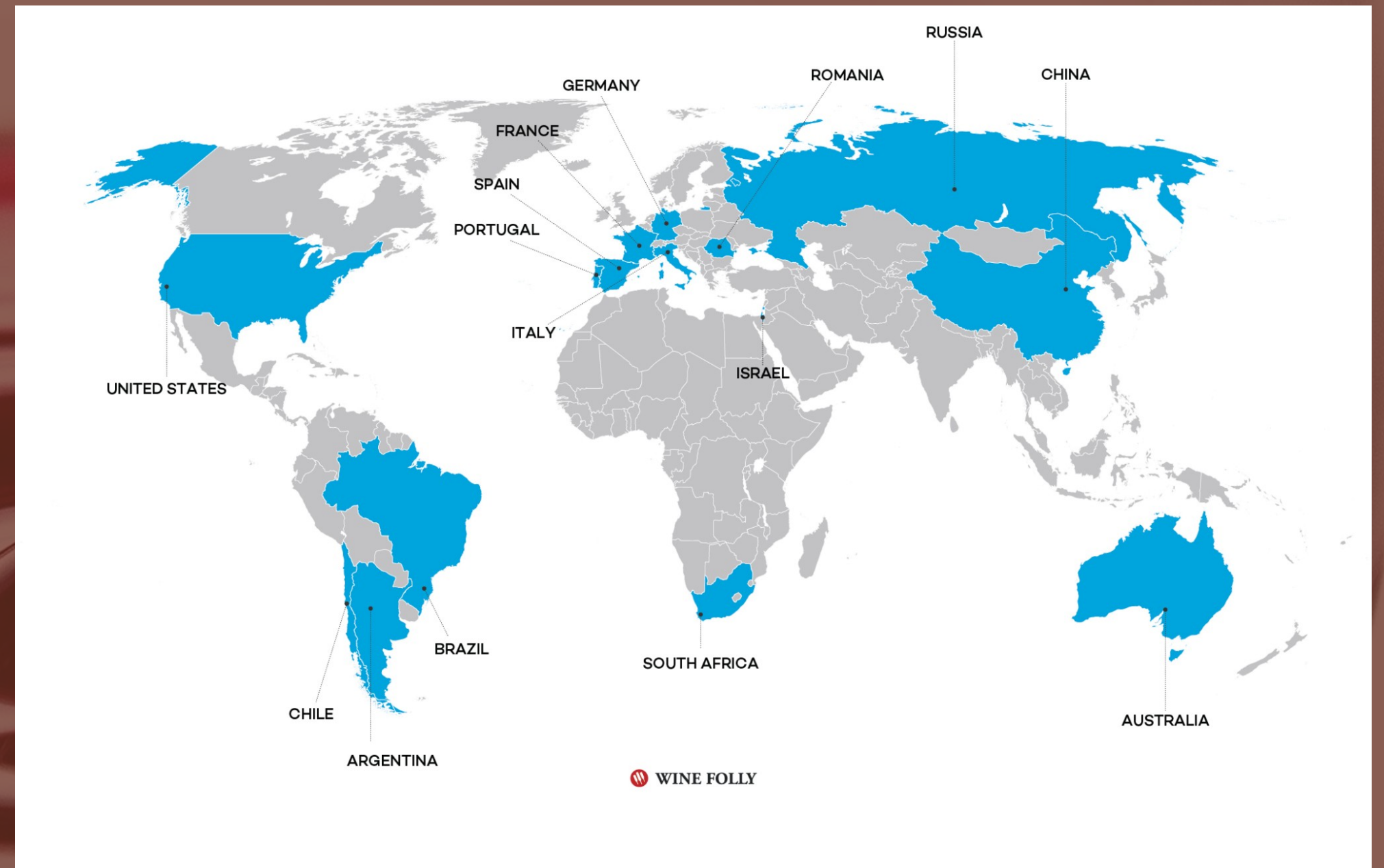
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Project Motivation & BackGround

“ Wine is one of the most
*culturally rich and globally
diverse beverages* “

Global Wine Map



Data Source Kaggle Link

 SAMUELMCGUIRE · UPDATED 4 YEARS AGO

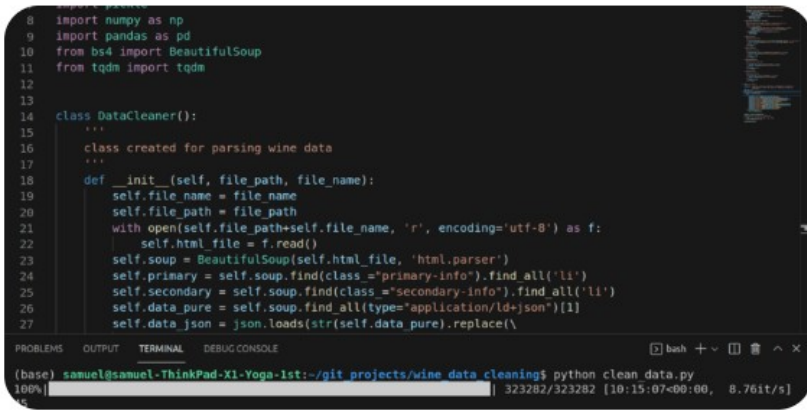
25

<> Code

Download

Wine Reviews Data

Dataset from scrape wine reviews



Data Card Code (5) Discussion (3) Suggestions (0)

About Dataset

Context

Part of a bigger project. See here for more information:
https://github.com/SamuelAdamsMcGuire/wine_data_clean

Usability ⓘ

8.24

License

Data files © Original Authors

Expected update frequency

Quarterly

323,231 Instances

11 Columns

- A multi-entity dataset with text, numeric, and categorical attributes.
- An ideal structure for relational modeling and analytical queries.

Data Extraction



Full Raw Data



6 Entities

Appellations

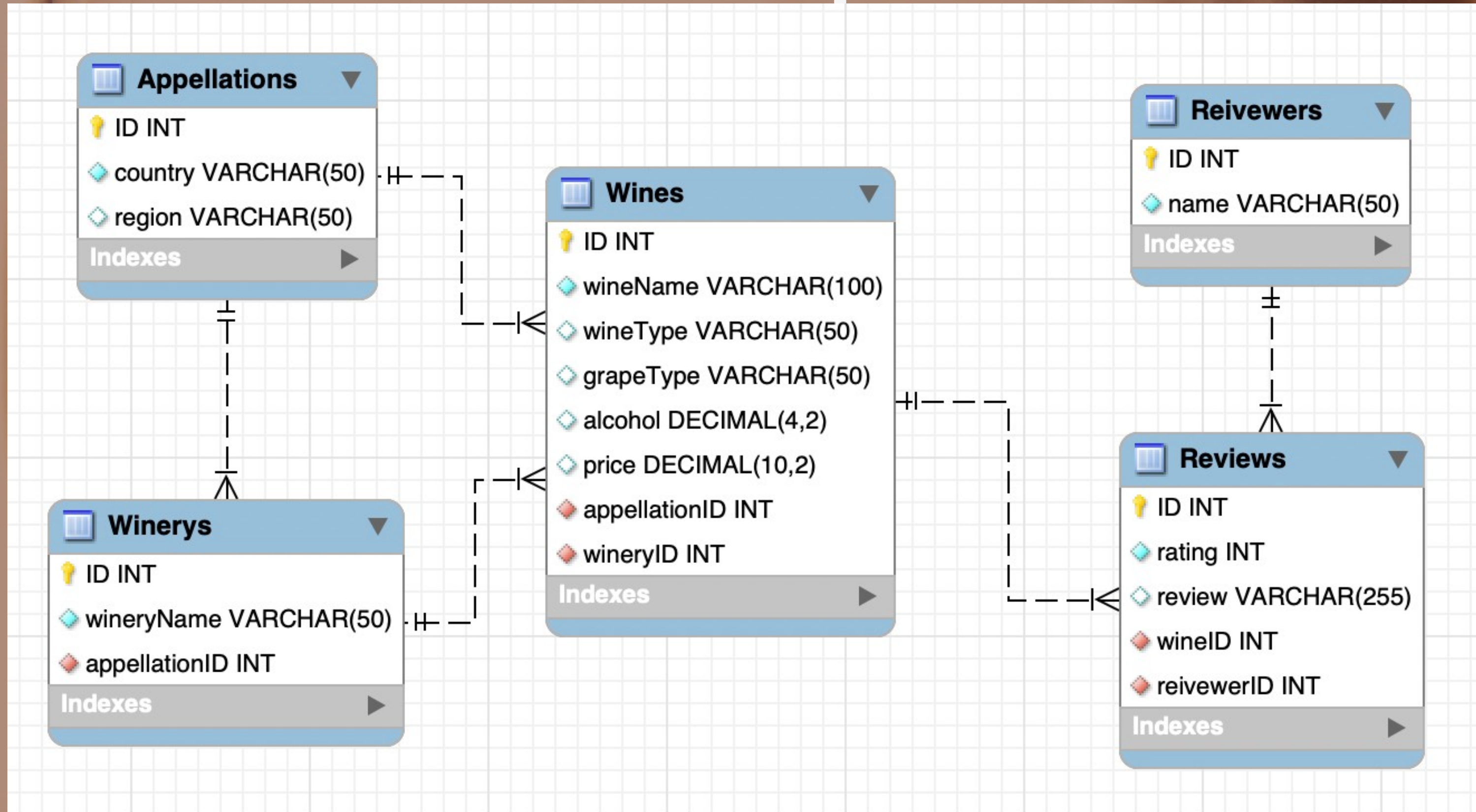
Winerys

Wines

Reviewers

Reviews

RDM - Add Relationships





Five Interesting Questions & SQL

SQL 1 Highest Rated Wines

**Q1) Which wines have the highest average rating,
considering only wines with at least 5 reviews?**

- This question helps identify the most consistently well-rated wines.
- It uses **grouping, aggregation, and filtering** to highlight top wines w/t sufficient review volume.

SQL 1 Highest Rated Wines



```
SELECT
    w.wineName,
    Round(Avg(r.rating), 2) AS avgRating,
    Count(r.ID) AS reviewCount
FROM
    Wines AS w
    INNER JOIN Reviews AS r ON w.ID = r.wineID
GROUP BY
    w.wineName
HAVING
    (((Count(r.ID)) >= 5))
ORDER BY
    Avg(r.rating) DESC,
    Count(r.ID) DESC;
```

wineName	avgRating	reviewCount
Chambers Rosewood Vineyards NV Rare Muscat (Rutherglen)	98.33	6
Chambers Rosewood Vineyards NV Grand Muscat (Rutherglen)	94.67	6
Lanson NV Extra Age Brut (Champagne)	93.6	5
Krug NV Brut Ros?? (Champagne)	92.83	6
Nino Franco NV Brut (Valdobbiadene Prosecco Superiore)	92.4	5
Charles Heidsieck NV Brut R??serve (Champagne)	92.33	6
Taittinger NV Pr??lude Grands Crus Brut (Champagne)	92.12	8
Pol Roger NV Pure Extra Brut (Champagne)	92	5
Laurent-Perrier NV Grand Si??cle La Cuv??e Brut (Champagne)	91.67	6
Krug NV Grande Cuv??e Brut (Champagne)	91.57	7
Billecart-Salmon NV Brut Ros?? (Champagne)	91.38	8
Bellavista NV Gran Cuv??e Sat??n (Franciacorta)	91.33	6
Bollinger NV Special Cuv??e Brut (Champagne)	91	10
Roederer Estate NV Brut Ros?? Sparkling (Anderson Valley)	91	10
Louis Roederer NV Brut Premier (Champagne)	91	9
Bodegas Dios Baco S.L. NV Amontillado Sherry (Jerez)	91	5
Chambers Rosewood Vineyards NV Muscat (Rutherglen)	90.83	6
Ruinart NV Blanc de Blancs Brut Chardonnay (Champagne)	90.71	7
Pol Roger NV R??serve Brut (Champagne)	90.62	8
Pol Roger NV Rich Demi-Sec (Champagne)	90.6	5
Bailly-Lapierre NV Brut (Cr??mant de Bourgogne)	90.4	10
Billecart-Salmon NV Brut R??serve (Champagne)	90.33	6
Lanson NV Rose Label Brut Ros?? (Champagne)	90.33	6
Vollereaux NV Blanc de Blancs Brut Chardonnay (Champagne)	90.29	7
Boizel NV Ultime Extra Brut (Champagne)	90.2	5
Montenisa NV Brut (Franciacorta)	90.17	6
Ayala NV Brut Majeur (Champagne)	90.17	6
Gosset NV Grande R??serve Brut (Champagne)	90.09	11
Taittinger NV La Fran??aise Brut (Champagne)	90	10
Mailly Grand Cru NV Brut R??serve (Champagne)	89.88	8
Boizel NV Blanc de Blancs Brut Chardonnay (Champagne)	89.8	5
Monte Rossa NV Prima Cuv??e Brut (Franciacorta)	89.8	5
Domaine Charles Baur NV Brut Sparkling (Cr??mant d'Alsace)	89.8	5
Jean Laurent NV Blanc de Noirs Brut Pinot Noir (Champagne)	89.8	5
Deutz NV Brut Classic (Champagne)	89.71	14

SQL 2 Country with Highest Average Rating

Q2) Which countries produce
the highest-rated wines on average?

- This query provides insights into which wine-producing countries demonstrate consistently high performance.
- It uses a **multi-table join** (Appellations → Wines → Reviews), making it a strong demonstration of relational understanding.

SQL 2 Country with Highest Average Rating



```
SELECT
  a.country,
  Round(Avg(r.rating), 2) AS avgRating,
  Count(r.ID) AS reviewCount
FROM
  (
    Appellations AS a
    INNER JOIN Wines AS w ON a.ID = w.appellationID
  )
  INNER JOIN Reviews AS r ON w.ID = r.wineID
GROUP BY
  a.country
HAVING
  (((Count(r.ID)) >= 10))
ORDER BY
  Avg(r.rating) DESC;
```

country	avgRating	reviewCount
Austria	90.53	7095
Buy Now	89.93	41
Germany	89.54	5554
France	89.38	57810
Hungary	89.14	333
Italy	88.84	42455
US	88.62	135246
Portugal	88.57	13299
Israel	88.32	1196
Australia	88.09	9594
New Zealand	87.95	4003
South Africa	87.91	3793
Greece	87.18	1155
Spain	87.14	14945
Argentina	86.59	8740
Chile	86.41	9534
Romania	85.98	331

finervs-Appellation

SQL 3 Best Value Wine

**Q3) Which wines offer the best value,
defined as an average rating of at least 95
and price under 30?**

- Consumers often seek wines with high ratings at affordable prices.
- This query combines **pricing** and **rating** data, demonstrating practical and domain-relevant analysis.

SQL 3 Best Value Wine

```
SELECT
    w.wineName,
    w.price,
    AVG(r.rating) AS avgRating,
    COUNT(r.ID) AS reviewCount
FROM
    Wines AS w
    INNER JOIN Reviews AS r ON w.ID = r.wineID
GROUP BY
    w.wineName,
    w.price
HAVING
    AVG(r.rating) >= 95
    AND w.price < "30"
ORDER BY
    AVG(r.rating) DESC,
    w.price ASC;
```

wineName	price	avgRating
T??rk 2016 Eiswein Gr??ner Veltliner (Nieder??sterreich)	\$25	97
T??rk 2018 Eiswein Gr??ner Veltliner (Nieder??sterreich)	\$25	97
Gritsch Mauritiushof 2019 Ried Steinporz Smaragd Gr??ner Veltliner (Wachau)	\$26	97
H.& C. Nittnaus 2015 Beerenauslese Exquisit White (Burgenland)	\$19	96
Domaine Andau 2017 Beerenauslese Scheurebe (Burgenland)	\$20	96
Rulo 2007 Syrah (Columbia Valley (WA))	\$20	96
Nikolaihof 2011 Vom Stein Federspiel Riesling (Wachau)	\$22	96
Piedrasassi 2018 Syrah (Santa Barbara County)	\$22	96
Brady Vineyard 2018 Zinfandel (Paso Robles)	\$24	96
Julien Schaal 2017 Sommerberg Granite Grand Cru Riesling (Alsace)	\$25	96
Julien Schaal 2019 Schiste Kastelberg Grand Cru Riesling (Alsace)	\$25	96
T??rk 2017 Eiswein Zweigelt (Nieder??sterreich)	\$25	96
J Wrigley 2017 Estate Riesling (McMinnville)	\$25	96
Zantho 2018 Trockenbeerenauslese White (Burgenland)	\$25	96
Isole e Olena 2010 Chianti Classico	\$27	96
Donkey & Goat 2012 Grenache Noir Grenache (El Dorado)	\$27	96
The Eyrie Vineyards 2014 Estate Chardonnay (Dundee Hills)	\$27	96
Weixelbaum 2016 Wahre Werte Wechselberg Himmel Auslese Sauvignon Blanc (Nieder??sterreich)	\$28	96
Ch??teau Filhot 2014 Sauternes	\$28	96
Eichinger 2014 Gaisberg Reserve Riesling (Kamptal)	\$28	96
Fess Parker 2008 Bien Nacido Vineyard Chardonnay (Santa Barbara County)	\$28	96
Palmento Costanzo 2016 Mofete Rosso (Etna)	\$29	96
Rosenhof 2015 Trockenbeerenauslese Chardonnay (Burgenland)	\$29	96
Adriano Marco & Vittorio 2016 Basarin (Barbaresco)	\$29	96
Hager Matthias 2018 Eiswein Zweigelt (Nieder??sterreich)	\$29	96
Domaines Schlumberger 2014 Saering Grand Cru Riesling (Alsace)	\$29	96
Apollonio 1997 Red (Copertino)	\$10	95
Santa Barbara Winery 2001 Lafond Vineyard Late Harvest Sauvignon Blanc (Santa Ynez Valley)	\$16	95
Saracco 2019 Moscato d'Asti	\$17	95
Lesehof Stag??rd 2016 Pfaffenberg Riesling (Kremstal)	\$20	95
Dunham 2010 Lewis Estate Vineyard Riesling (Columbia Valley (WA))	\$20	95
Styring 2014 Whimsy Estate Riesling (Ribbon Ridge)	\$20	95
Winzerkeller Andau 2017 Beerenauslese Scheurebe (Burgenland)	\$20	95
Januik 2012 Bacchus Vineyard Riesling (Columbia Valley (WA))	\$20	95
Daltrineri 2019 Radice (Lambrusco di Sorbara)	\$20	95

SQL 4 Winery Concentration by Appellation

Q4) How many wineries are located in each appellation (country and region), and which region exhibits the highest concentration of producers?

- This question uses the **one-to-many relationship** between Appellations and Wineries to show **how producer density varies** by region.
- It applies a simple **multi-table join** (Appellations → Wineries) with COUNT to highlight regional specialization.

SQL 4 Winery Concentration by Appellation



```
SELECT
    A.country,
    A.region,
    COUNT(W.ID) AS WineryCount
FROM
    Appellation AS A
    INNER JOIN Winerys AS W ON A.ID = W.appellationID
GROUP BY
    A.country,
    A.region
ORDER BY
    COUNT(W.ID) DESC;
```

country	region	WineryCount
US	Napa Valley, Napa, California	1450
US	California, California Other, California	1338
US	Russian River Valley, Sonoma, California	775
US	Columbia Valley (WA), Columbia Valley, Wasl	687
US	Paso Robles, Central Coast, California	631
Argentina	Mendoza, Mendoza Province	624
US	Sonoma County, Sonoma, California	583
US	Sonoma Coast, Sonoma, California	528
France	Bordeaux, Bordeaux	498
US	Willamette Valley, Willamette Valley, Oregon	479
Italy	Toscana, Tuscany	479
US	Central Coast, Central Coast, California	460
US	Santa Barbara County, Central Coast, Californ	448
Italy	Barolo, Piedmont	386
Spain	Rioja, Northern Spain	383
US	Carneros, Napa-Sonoma, California	379
France	Bordeaux Blanc, Bordeaux	374
France	Bordeaux Sup??rieur, Bordeaux	365
US	Lodi, Central Valley, California	358
US	Dry Creek Valley, Sonoma, California	355
New Zealand	Marlborough	328
Italy	Chianti Classico, Tuscany	326
France	Saint-??milion, Bordeaux	324
France	Champagne, Champagne	305
Italy	Brunello di Montalcino, Tuscany	302
France	C??tes de Provence, Provence	301
France	C??tes du Rh??ne, Rh??ne Valley	297
Spain	Ribera del Duero, Northern Spain	293
US	Sta. Rita Hills, Central Coast, California	289
US	Alexander Valley, Sonoma, California	283

SQL 5 Price Differences Across Appellation

Q5) How do average wine prices differ across appellations (country and region)?

- This question examines price differences across appellations, highlighting which regions tend to produce more premium or budget-friendly wines.
- It relies on the Appellations → Wines relationship, using aggregation to compare regional pricing patterns.

SQL 5 Price Differences Across Appellation

```
SELECT
  a.country,
  a.region,
  Round(AVG(w.price), 2) AS avgPrice,
  COUNT(w.ID) AS wineCount
FROM
  Appellations AS a
  INNER JOIN Wines AS w ON a.ID = w.appellationID
WHERE
  w.price IS NOT NULL
GROUP BY
  a.country,
  a.region
HAVING
  COUNT(w.ID) >= 5
ORDER BY
  a.country,
  AVG(w.price) DESC;
```

country	region	avgPrice	wineCount
Argentina	Perdriel, Mendoza Province	77.33	49
Argentina	R??o Negro Valley, Other	55.59	61
Argentina	La Consulta, Mendoza Province	42.96	92
Argentina	Paraje Altamira, Mendoza Province	42.76	25
Argentina	Los Chacayes, Mendoza Province	38.67	9
Argentina	Gualtallary, Mendoza Province	37.92	12
Argentina	Luj??n de Cuyo, Mendoza Province	30.29	370
Argentina	Pedernal Valley, Other	28.5	10
Argentina	Uco Valley, Mendoza Province	28.43	737
Argentina	Agrelo, Mendoza Province	27.75	122
Argentina	Tupungato, Mendoza Province	24.2	411
Argentina	Calchaqu?? Valley, Other	23.74	86
Argentina	Salta, Other	21.98	148
Argentina	Mendoza, Mendoza Province	21.4	4963
Argentina	San Carlos, Mendoza Province	21.11	18
Argentina	Patagonia, Other	21.01	115
Argentina	Vista Flores, Mendoza Province	20.36	39
Argentina	Alto Valle del R??o Negro, Other	19.57	37
Argentina	Cafayate, Other	19.46	222
Argentina	Argentina, Other	19.31	103
Argentina	Mendoza Province	18.86	139
Argentina	Maip??, Mendoza Province	18.23	53
Argentina	Neuqu??n, Other	17.48	83
Argentina	Zonda Valley, Other	15.67	6
Argentina	Famatina Valley, Other	13.95	38
Argentina	San Juan, Other	13.82	158
Argentina	Aminga Valley, Other	13.45	11
Argentina	Tulum Valley, Other	12.57	35
Argentina	La Rioja, Other	11.74	42



Forms

Main Form - Appellations

MainForm_Appellations

ID

country

Argentina

region

Agrelo, Mendoza Province

subWinerys

	ID	wineryName	appellationII			
+	1021	Bodega Catena Zapata	1			
+	1022	Bodegas Escorihuela	1			
+	1023	Cabra Montes	1			
+	1024	Cadus	1			
+	1025	Casarena	1			
+	1026	El Enemigo	1			
+	1027	Finca Decero	1			
+	1028	Finca La Anita	1			
+	1029	High Altitude	1			

Record:

1 of 18

No Filter

Search

- **Role:** This is the highest-level Main Form used for managing geographical information, specifically the country and region.
- **Data Relationship:** This form can contain multiple wineries (it represents the '1' side of a 1:M relationship).
- **Typical Usage:** This form is used as the Main Form; the Winerys Form is inserted as a Subform to display a list of all wineries located within the selected region.

Main Form/Subform - Winerys

SubForm_Winerys

ID

25

wineryName

Artesana

appellationID

subWines

wineName	wineType	grapeType	alcohol	price	appellatic	wineryID
Artesana 2010 Estate Blend Tannat-Merlot (Canelones)	Red		0.14	\$23		25
Artesana 2011 Tannat (Canelones)	Red	Tannat, Other Red	0.15	\$29		25
Artesana 2011 Tannat-Merlot (Canelones)	Red		0.153	\$29		25
Artesana 2012 Tannat (Canelones)	Red	Tannat, Other Red	0.139	\$16		25
Artesana 2012 Tannat-Merlot (Canelones)	Red		0.139	\$16		25
Artesana 2013 Tannat (Canelones)	Red	Tannat, Other Red	0.149	\$16		25
Artesana 2013 Tannat-Merlot-Zinfandel Red (Canelones)	Red	Red Blends, Red Blends	0.145	\$16		25
Artesana 2015 Tannat (Canelones)	Red	Tannat, Other Red	0.138	\$18		25
Artesana 2015 Tannat-Merlot-Zinfandel Red (Canelones)	Red	Red Blends, Red Blends	0.138	\$18		25
Artesana 2016 Reserva Tannat (Canelones)	Red	Tannat, Other Red		\$19		25
Artesana 2016 Tannat Ros?? (Canelones)	Rose	Ros??	0.13	\$14		25
Artesana 2017 Reserva Cabernet Franc-Merlot (Canelones)	Red	Cabernet Franc-Merlot, Borde		\$19		25
Artesana 2017 Reserva Tannat (Canelones)	Red	Tannat, Other Red	0.142	\$19		25
Artesana 2017 Tannat Ros?? (Canelones)	Rose	Ros??	0.135	\$14		25
Artesana 2017 Tannat-Merlot-Zinfandel Red (Canelones)	Red	Red Blends, Red Blends		\$19		25
Artesana 2018 Reserva Cabernet Franc-Merlot (Canelones)	Red	Cabernet Franc-Merlot, Borde	0.14	\$19		25
Artesana 2018 Reserva Tannat (Canelones)	Red	Tannat	0.15	\$19		25
Artesana 2019 Tannat Ros?? (Canelones)	Rose	Ros??	0.135	\$14		25
Artesana 2020 Tannat Ros?? (Canelones)	Rose	Ros??	0.135	\$14		25

Record: 1 of 19

No Filter

Search

- **Role:** This is a Main Form used for managing producer information, such as the winery Name and other details.
- **Data Relationship:** This form can produce multiple wines (it represents the '1' side of a 1:M relationship).
- **Typical Usage:** This form is used as the Main Form; the Wines Form is inserted as a Subform to display the list of wines produced by the selected winery.

Sub Form - Wines

SubForm_Wines

ID

1

wineName

Črnko 2012 Jarenincan White (??tajerska)

wineType

White

grapeType

White Blend

alcohol

0.11

price

\$15

appellationID

wineryID

1

- **Role:** This is a detail data form used for managing specific product information, such as the wine's name, price, and vintage details.
- **Data Relationship:** It is primarily used as a Subform within the Winerys Form, displaying a list of wines related to a specific winery (the 'M' side). It also forms an M:M relationship with the Reviews table.



Report

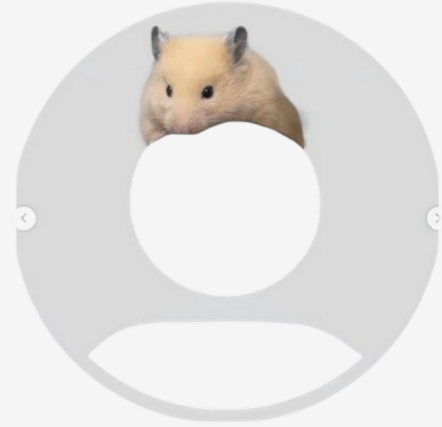
Price Differences Across Appellations Report

PriceDifferencesAcrossAppellatio			
country	region	avgPrice	ineCount
Argentina			
	Agrelo, Mendoza Province	27.75	122
	Alto Valle de Uco, Mendoza Province	11.67	6
	Alto Valle del R??o Negro, Other	19.57	37
	Aminga Valley, Other	13.45	11
	Argentina, Other	19.31	103
	Cafayate, Other	19.46	222
	Calchaqu?? Valley, Other	23.74	86
	Famatina Valley, Other	13.95	38
	Gualtallary, Mendoza Province	37.92	12
	La Consulta, Mendoza Province	42.96	92
	La Rioja, Other	11.74	42
	Los Chacayes, Mendoza Province	38.67	9
	Luj??n de Cuyo, Mendoza Province	30.29	370
	Maip??, Mendoza Province	18.23	53
	Mendoza Province	18.86	139
	Mendoza, Mendoza Province	21.4	4963
	Neuqu??n, Other	17.48	83
	Paraje Altamira, Mendoza Province	42.76	25
	Patagonia, Other	21.01	115
	Pedernal Valley, Other	28.5	10
	Perdriel, Mendoza Province	77.33	49
	R??o Negro Valley, Other	55.59	61
	Salta, Other	21.98	148
	San Carlos, Mendoza Province	21.11	18
	San Juan, Other	13.82	158
	Tulum Valley, Other	12.57	35
	Tupungato, Mendoza Province	24.2	411
	Uco Valley, Mendoza Province	28.43	737
	Vista Flores, Mendoza Province	20.36	39
	Zonda Valley, Other	15.67	6
Australia			
	Adelaide Hills, South Australia	26.46	355
	Adelaide Plains, South Australia	13.4	5
	Adelaide, South Australia	25.24	58
	Alpine Valleys, Victoria	12.6	5
	Australia, Australia Other	38.6	87
	Barossa Valley, South Australia	44.11	876

This report is designed to analyze wine pricing patterns by showing the average price and volume of wine listings across various Appellations (countries and regions).

- **country:** Organizes the report into logical country sections for focused analysis.
- **region (Ascending Order):** Within each country, regions are listed in alphabetical order for quick look-up.
- **avgPrice (Descending Order):** Within each region, records are ordered from the highest average price down, immediately identifying the most premium market segments.
- **wineCount:** Where prices are identical, the records with the highest volume of listings WineCount are shown first, lending greater statistical context to the price data.

Individual Contributions



Jaehee Jung

- Topic and dataset selection
- Project statement development
- Data preprocessing and entity extraction
- Designing 5th SQL question & create query
- Creating 3rd Form in MS Access
- Creating Report in MS Access
- PowerPoint completion



Donghun Yoo

- Data preprocessing and entity extraction
- Designing 3rd, 4th SQL question & create query
- Creating 2nd Form in MS Access
- Creating Report in MS Access
- PowerPoint completion



Dongchan Kim

- Project statement development
- Data preprocessing and entity extraction
- Designing 1st, 2nd SQL question & create query
- Creating 1st Form in MS Access



THANK
YOU